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DEPARTMENT OF GOVERNMENT EXAMINATIONS – CHENNAI-6
HSC SECOND YEAR EXAMINATION MARCH - 2024
CHEMISTRY ANSWER KEY

Note: 1. Answer written with Blue or Black ink only to be evaluated

2. Choose the most suitable answer in **PART – I** from the given alternatives and write the option code and the corresponding answer.

Maximum Marks: 70

PART- I

Answer all the questions **15×1=15**

Q.No	Option Code	'A' Type	Q. No	Option Code	'B' Type
1	b)	Both Assertion and Reason are true and Reason is the correct explanation of Assertion	1	b)	First order
2	c)	Potassium trioxalato aluminate (III)	2	b)	Acetyl salicylic acid
3	b)	HI	3	d)	carbon dioxide
4	c)	Dry ice	4	c)	Potassium trioxalato aluminate (III)
5	c)	Cytosine and Uracil	5	a)	Sodium chloride
6	b)	Acetyl salicylic acid	6	b)	(i) and (iv)
7	d)	carbon dioxide	7	d)	Impure copper
8	c)	acetanilide	8	b)	Both Assertion and Reason are true and Reason is the correct explanation of Assertion
9	a)	Sodium chloride	9	c)	Nucleophilic addition
10	b)	(i) and (iv)	10	c)	Dry ice
11	c)	Nucleophilic addition	11	d)	PCC
12	b)	First order	12	b)	HI
13	d)	Impure copper	13	c)	acetanilide
14	c)	Al_2O_3	14	c)	Cytosine and Uracil
15	d)	PCC	15	c)	Al_2O_3

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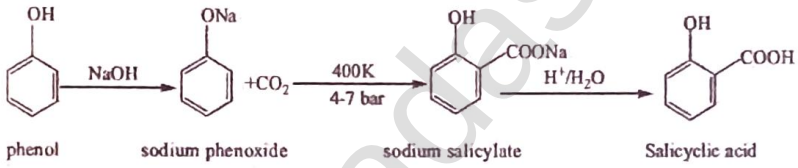
Answer any SIX Questions and Question **No.24 is Compulsory.** **6×2=12**

Q.No	Answer	Marks	
16	Correct Explanation **(or)** Correct Equation	2	2
17	Correct equation **(or)** unbalanced equation **(or)** equation without condition **(or)** mere explanation alone	2 1½ 1	2
18	Any one correct equation **(or)** Explanation (Any one)	2 1	2
19	Correct explanation Example	1 1	2
20	$E^{\circ}_{cell} = (E^{\circ}_{oxi}) + (E^{\circ}_{red})$ **(or)** $= -1.09+0.771$ $E^{\circ}_{cell} = -0.319$ V **(or)** E°_{cell} is –ve , Fe^{3+} cannot oxidises Br^{-} to Br_2	½ ½ 1	2
21	 **(or)** equation without condition **(or)** explanation only	2 1	2
22		1+1	2
23	Correct explanation **(or)** one example	2 1	2
24	correct definition	2	2

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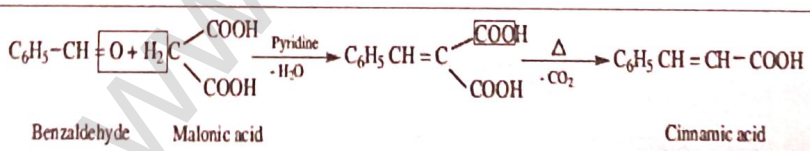
Part–III

Answer any **SIX** Questions and **Question No.33 is Compulsory.** 6×3=18

Q.No	Answer	Marks	
25	Any three uses	3	3
26	Fe^{3+}= [Ar] $3d^5$ Fe^{3+} - half-filled d orbital Fe^{3+} is stable	1 1 1	3
27	$r = \frac{a\sqrt{2}}{4}$ **(or)** $a = \frac{4r}{\sqrt{2}}$ **(or)** $a = 2\sqrt{2}\,r$ **(or)** a=2x1.414 r $a = \frac{4 \times 125}{\sqrt{2}}$ **(or)** $a = 2\sqrt{2}\,125$ **(or)** a=2x1.414 x125 a = 353.5 pm	1 1 ½+½	3
28	$K = Ae^{-Ea/RT}$ K = Rate constant A = Frequency factor R = Gas constant E_a = Energy of activation T =Temperature	1 4x½	3
29	**Physisorption** / **chemisorption** Physisorption decreases with increase in temperature. / When temperature is raised chemisorption first increases and then decreases. In Physisorption, when pressure increases the extent of adsorption increases. / Chemical adsorption is fast with increase Pressure, it cannot alter the amount.	1½ 1½	3
30	 Benzaldehyde Malonic acid Cinnamic acid **(or)** equation without condition **(or)** explanation only	3 2½ 1	3

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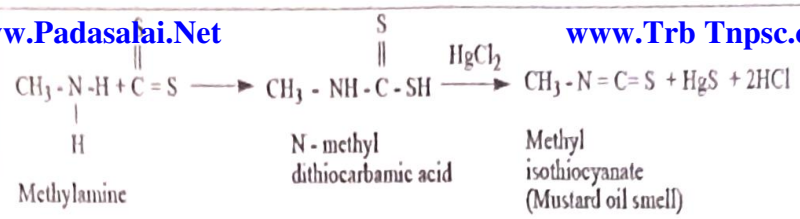
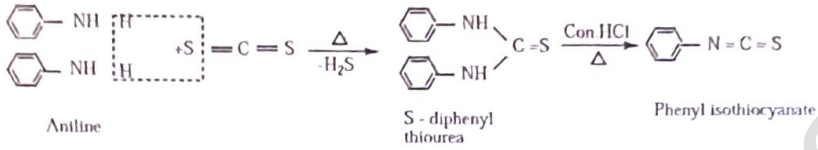
31	 Methylamine N - methyl dithiocarbamic acid Methyl isothiocyanate (Mustard oil smell) (or)  Aniline S - diphenyl thiourea Phenyl isothiocyanate **(or)** equation without condition **(or)** explanation only	3 2½ 1	3
32	Correct explanation **(or)** correct example **(or)** -CO-NH- bond only	3 1	3
33	Mere Attempt	3	3

Part– IV

Answer all the Questions **5x5=25**

Q.No	Answer	Marks	
34 (a)	i) Any 3 differences	3	5
	ii) Silica acts as a flux **(or)** Correct Equation only	2	
	(OR)		
(b)	i) Any 3 uses	3	5
	ii) Correct definition **(or)** Correct Structure only	2 1	
35 (a)	Lanthanoid contraction – Explanation Lanthanoid contraction consequences **(Any Three)**	2 3	5
	(OR)		
(b)	i) Double salt – Explanation **(or)** Double salt – Example Co-ordination Compound – Explanation **(or)** Co-ordination Compound – Example	1½ 1 1½ 1	5
	ii) One example for medicinal importance One example for biological importance	1 1	

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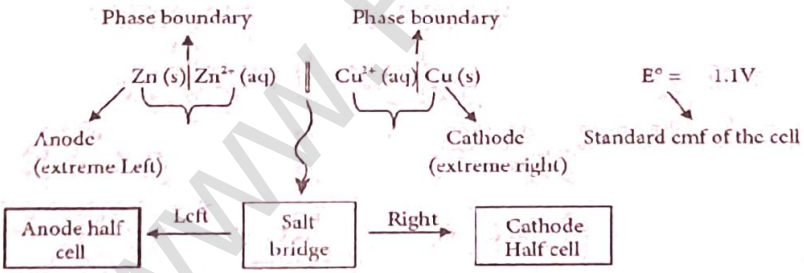
36 (a)	Packing efficiency = $\frac{\text{total volume occupied by spheres in a unit cell}}{\text{volume of the unit cell}} \times 100$	1	
	Volume of cube = a x a x a = a^3		
	(or)	1	
	Radius of the sphere from figure, a = 2r(or)r = $\frac{a}{2}$		
	Volume of the sphere with radius = $\frac{4}{3}\pi\left(\frac{a}{2}\right)^3$	1	5
	(or) $= \frac{\pi a^3}{6}$		
	Packing efficiency = $\frac{1\times\frac{\pi a^3}{6}}{a^3}$ x 100	1	
	= 52.38%	1	
	(OR)		
(b)	i) Rate = $k[A]^0$ **(or)** $\frac{-d[A]}{dt}$ = k **(or)** - d[A]= kdt	1	
	$-\int_{[A_0]}^{[A]} d[A] = K\int_0^t dt$ **(or)** $-([A])_{[A_0]}^{[A]} = k\,(t)_0^t$	1	5
	$[A_0] - [A]$= kt **(or)** k = $\frac{[A_0]-[A]}{t}$	1	
	ii) Correct definition **(or)**	2	
	Formula	1	
37 (a)	i) 	3	5
	(or)		
	Correct Explanation	3	
	(or)		
	Zn | Zn^{2+} || Cu^{2+} | Cu $E^0 = 1.1V$	1	
	ii) Correct definition	2	

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(b)	$CH_3-C(CH_3)_2-OH + HCl \xrightarrow{\text{anhydrous } ZnCl_2} CH_3-C(CH_3)_2-Cl + H_2O$ 2-methylpropan-2-ol → 2-chloro-2-methylpropane (immediate appearance of turbidity)	2	5
	$CH_3-CH(OH)-CH_3 + HCl \xrightarrow{\text{anhydrous } ZnCl_2} CH_3-CH(CH_3)-Cl + H_2O$ propan-2-ol → 2-chloropropane (slow appearance of turbidity)	2	
	$CH_3-CH_2-OH + HCl \xrightarrow{\text{anhydrous } ZnCl_2}$ No reaction at room temperature ethanol (Turbidity appears only on heating)	1	
	(or) Correct equation without mentioning turbidity	3	
	(or) Correct equation without mentioning Lucas reagent	3	
38 (a)	i)Any one correct equation	2	5
	(or)		
	Correct Explanation	1	
	ii)Correct definition	2	
	Any one example	1	
	(OR)		
(b)	$CH_3-\underset{\underset{O}{\|}}{C}-H + 4(H) \xrightarrow[\text{Con HCl}]{\text{Zn - Hg}} CH_3-CH_3 + H_2O$ Acetaldehyde **(A)** → Ethane **(B)**	2	5

Compound	Formula		Name
A	CH_3CHO	(or)	Acetaldehyde
B	CH_3CH_3		Ethane
E	CH_3COOH		Acetic acid

3×1

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